

Examine how proposed scientific responses to contemporary issues may impact on society and explore ethical, environmental, social and economic considerations

Learning intention

- examine how proposed scientific responses to contemporary issues may impact on society and explore ethical, environmental, social and economic considerations.

Teach and model

- discussing how scientific knowledge of the forces involved in flight has led to changes in aircraft design and any ethical, environmental.
- examining how the use of desalination plants to produce fresh water has impacted marine ecosystems where the desalination plants are located.
- investigating how scientific knowledge that larger reserves are better for maintaining ecosystem function might interact with competing viewpoints.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.